

# **"Jianggang Plan" Desert Ecological Restoration Application — A Theoretical Feasibility Study Report**

**Project Title: Ecological Restoration Application of Multi-source Solid Waste Synergistic Transformation Technology Based on Accelerated Artificial Diagenesis in Desertification Environments**

**Application Scenario: Middle East (Oman, UAE, Saudi Arabia) and Africa (Mauritania, Sahara/Sahel Region)**

**Research Type: Theoretical Feasibility Study**

## **I. Core Concept: Why Can the "Jianggang Plan" Fill the Desert Ecological Niche?**

**The core deficiency of desert ecosystems lies in three structural defects: lack of water, lack of organic matter, and lack of soil structure. Traditional desertification control methods — physical sand barriers, chemical sand-fixing agents, and long-distance water diversion — each have their merits, but all suffer from limitations such as high cost, poor sustainability, or secondary pollution.**

**The "Jianggang Plan" offers a fundamentally different technical pathway: using local industrial solid waste (steel slag) as the inorganic skeleton, and local organic waste (agricultural residues, municipal organic waste) as biochemical fuel, to construct an "artificial diagenesis" ecological reactor that "synthesizes" functional soil matrix in situ within the desert. The essence of this concept is to transform the desert from a "lifeless mineral accumulation" into an "ecosystem substrate with biological activity and water-holding capacity."**

The core logic chain of this approach is clear: steel slag provides the mineral skeleton (Ca, Si, Mg, Fe, etc.) and alkaline reaction sites; organic waste provides the carbon source, microbial energy, and water-retaining substrate; the MICP functional microbial community drives carbonate cementation and soil aggregate formation; and saprophagous insects such as black soldier flies accelerate organic matter conversion while producing

nutrient-rich larval castings — these four components work synergistically to construct a self-sustaining "artificial ecological niche" in the desert environment.

## **II. Needs Analysis: The Urgency of Ecological Restoration in the Middle East and Africa**

### **2.1 Middle East Region**

**The Middle East and North Africa (MENA) region is one of the areas most severely affected by land degradation and desertification worldwide. Large tracts of land in the MENA region have been severely degraded due to multiple factors including climate change and saline irrigation. In the Sultanate of Oman, approximately 44% of agricultural land has been affected by salinization, most severely in North Al Batinah and South Al Batinah, directly threatening crop yields and national food security. Gulf countries such as the UAE and Saudi Arabia similarly face severe desertification challenges, with extremely low soil organic matter content (typically below 0.5%) and virtually zero water and nutrient retention capacity.**

### **2.2 Africa Region**

**The sub-Saharan Sahel region is a global focal point for desertification control. The Food and Agriculture Organization (FAO) of the United Nations has secured US\$222 million in support from the Green Climate Fund (GCF) to launch the SURAGGWA program, promoting low-emission, climate-resilient land management practices across the Sahel, with a target of restoring over 1.2 million hectares of degraded land. In Mauritania, the Xinjiang Institute of Ecology and Geography, Chinese Academy of Sciences, has collaborated with Mauritanian partners to successfully complete sand fixation on 100 hectares of sandy land, of which 4 hectares have been converted into sustainably utilized oases, and 45 local technicians have completed greening technology training. Mauritania's Minister of the Environment has described China as an "irreplaceable" scientific and technological partner for Mauritania.**

### **III. Resource Endowment: Industrial Solid Waste and Organic Waste Stocks in Oman and Target Regions**

A core prerequisite of the "Jianggang Plan" is that target regions must simultaneously possess sources of steel slag and organic waste. Current data indicate that Oman fully meets this condition, while African target regions are also developing rapidly.

#### **3.1 Oman: Ready for Launch**

**Steel Slag Resources.** Oman possesses a mature and rapidly growing steel industry. Jindal Shadeed Iron and Steel (JSIS) operates large-scale steel production facilities at Sohar Port, and held a dedicated promotional event in January 2025 showcasing the sustainable application of steel slag in road construction and infrastructure projects. JSIS CEO Harssha Shetty has explicitly stated that steel slag utilization helps support Oman's environmental objectives. Furthermore, Oman is emerging as a global green steel hub — Jinnan Steel Group, in partnership with Zhongye Changtian, is constructing a green beneficiation project at Sohar, Oman, with a planned annual processing capacity of 18 million tons of iron ore and an annual output of 12.6 million tons of ultra-high-quality iron concentrate, providing a "Chinese solution" for the global steel industry. Merani Green Steel Company also plans to build a green steel production base in Oman. Once these projects come online, Oman's annual steel slag output will increase substantially, providing ample raw materials for the "Jianggang Plan."

**Organic Waste.** Oman has launched its first waste-to-energy project at Barka, with a total investment of US\$1 billion, processing 3,000 tons of municipal solid waste per day and generating approximately 757 GWh of electricity annually. In this municipal solid waste stream, the organic fraction (food waste, etc.) typically accounts for 40–60%, yielding approximately 1,200–1,800 tons per day of organic waste available for the "Jianggang Plan." In addition, Oman's agricultural waste (date palm branches, crop residues, etc.) is also abundant. Research teams at Sultan Qaboos University (SQU) have successfully produced biochar from date palm waste biomass and achieved significant results in saline-alkali soil improvement.

**Desert Land Resources.** Oman's Vision 2040 has positioned environmental sustainability as a core pillar of national strategy. The "Jianggang Plan" can be deeply integrated with large-scale ecological restoration initiatives such as Oman's "Plant Oman 2050."

### **3.2 African Target Regions**

Africa also possesses foundational resource conditions. Mauritania already has a cooperative foundation for desertification control with the Chinese Academy of Sciences. The FAO's SURAGGWA program, covering multiple Sahel countries, provides a policy and funding support framework. Several African countries have steel industries (South Africa, Egypt, Nigeria, etc.) with available steel slag resources; agricultural waste (such as date palm residues and crop straw) and municipal organic waste are also abundant.

### **3.3 Overall Resource Matching Assessment**

## **IV. Theoretical Feasibility: Applicability of Each Technical Module in Desert Environments**

The core technical modules of the "Jianggang Plan" — steel slag soil improvement, biochar water retention, MICP sand fixation, and organic waste conversion — have been supported by extensive pilot studies demonstrating solid theoretical foundations for their applicability in arid/desert environments.

### **4.1 Module 1: Theoretical Basis of Steel Slag as a Desert Soil Amendment**

The application of steel slag in agriculture and soil improvement is already a mature research field. Multiple significant recent studies have validated the exceptional potential of steel slag as a soil amendment.

**Nutrient Supply and Soil Chemical Regulation.** A review published in November 2025 systematically summarized the application of LD steel slag in sustainable agriculture: steel slag is rich in plant-essential nutrients — calcium, silicon, magnesium, phosphorus, and micronutrients including iron and manganese; its alkaline characteristics effectively neutralize soil acidity, improve nutrient

availability, and support soil microbial activity. Combined use of steel slag and biochar increased paddy soil pH by 14.2% and salinity by 98% (compared to untreated soil). Field studies in Bangladesh, China, Japan, Indonesia, and South Korea have shown that steel slag fertilizer can reduce methane emissions from paddy fields by 0.6% to 56%, depending on soil type, steel slag composition, and application rate. When combined with materials such as biochar, steel slag enhances carbon retention and microbial health, supporting long-term soil fertility.

**Crop Yield Effects.** A Korean study published in February 2026 systematically evaluated the effects of steel slag application in rice cultivation systems. Results showed that steel slag application significantly raised soil pH from 6.6 to 7.5, effectively neutralizing soil acidity and creating a more favorable growing environment for crops; steel slag significantly improved  $\text{SiO}_2$  availability, with mature grain weight peaking at 29.95 g under an application rate of 8 t/ha. Critically, heavy metals (Cr, Cd, Pb, As) were not detected in any of the treated soil or rice components, alleviating immediate toxicity concerns.

**Physical and Mechanical Improvement of Sandy Soils.** A 2025 study published in *Case Studies in Construction Materials* systematically investigated the combined use of steel slag and biopolymer cellulose for stabilizing sandy soils, validating the applicability of steel slag as an engineering amendment material in desert sandy soil environments.

**Comprehensive Remediation of Saline-Alkali Soils.** A study published in July 2025, based on circular economy principles, discussed an integrated system linking steel slag desulfurization with saline-alkali soil management, demonstrating that steel slag can achieve near-perfect desulfurization efficiency while its byproducts significantly enhance soil quality.

**Integrated Water and Nutrient Management Under Drought Conditions.** A recent study (November 2025) applied steel slag-based compost tea to lettuce cultivation under drought conditions, showing that nutrients in steel slag — calcium,

magnesium, silicon — improved soil structure and promoted plant growth. Researchers observed significantly enhanced drought resistance and yield in lettuce.

**Heavy Metal Risk Assessment and Control.** A major 2025 review critically discussed the challenges, scope, and opportunities of steel slag utilization in agriculture, noting that Japan, the United States, and European countries have taken the lead in developing steel slag-based fertilizers/amendments leveraging steel slag's nutrient-rich properties. However, the presence of heavy metals (particularly chromium) remains the primary challenge limiting its agricultural application. The "Jianggang Plan" has already designed a comprehensive self-stabilization mechanism for heavy metal risk control — heavy metals are activated and leached during the low-pH Phase 1, then re-immobilized in Phases 2–3 as pH recovers, through a triple pathway of co-precipitation, C-S-H gel adsorption, and biochar complexation. Monthly sampling monitors key heavy metal concentrations (Cr(VI), Ni, Pb, etc.) in leachate to ensure controllable risk management.

#### **4.2 Module 2: Water and Nutrient Retention Effects of Biochar in Desert Soils**

Biochar is a key technical means for addressing the water retention problem of desert soils. Multiple recent 2025 studies have validated the exceptional efficacy of biochar in sandy and semi-arid soils.

In sandy soil experiments under semi-arid conditions in northeastern Brazil, sewage sludge biochar (SSB) applied at 20 t/ha significantly increased soil organic carbon, nitrogen content, water holding capacity, and microbial biomass. Water use efficiency (WUE) improved by 148%, grain yield increased by 146% compared to conventional NPK fertilization, while toxic elements in legume grains remained within safe levels. Another study found that even low application rates of biochar improved the quality and microbial activity of barren desert sandy soil, benefiting mung bean growth and soil water retention.

In the Middle East, biochar produced from date palm residues through slow pyrolysis at 500°C had a moisture content as low as 3.52%, indicating its capacity to absorb and retain water when applied to soil, thereby enhancing soil water retention

capacity. In Oman itself, the SQU research team applied sulfur-treated biochar combined with organic compost to saline-alkali soil, achieving up to a 200% improvement in plant growth and soil fertility, with direct application technique refinements further enhancing the effect by an additional 50%.

#### **4.3 Module 3: Breakthrough Progress in MICP Technology for Desert Sand Fixation**

MICP technology provides a revolutionary means for desert sand fixation and soil formation. Recent research findings are encouraging.

**Mechanisms and Parameter Optimization for Desert Sand Fixation.** A cutting-edge study published in August 2025 (indexed in ScienceDirect) systematically investigated the sand fixation mechanism of MICP technology through a multi-scale approach from macro to micro. The study employed Response Surface Methodology (RSM) to optimize process parameters: optimal cementation solution concentration of 1.895 mol/L, with an optimal treatment cycle of 4 rounds. The optimized parameters showed good agreement with experimental results, with a deviation of less than 5% between repeated tests and the optimized outcome. Under both high-concentration single treatment and low-concentration multiple treatment conditions, MICP-treated desert sand achieved efficient sand fixation and durable water retention. Microanalysis revealed a clear correlation among  $\text{CaCO}_3$  content, solidified layer thickness, and surface strength. MICP transformed the contact mode between sand particles from "point contact" to "surface contact," with a significant negative correlation between pore parameters and surface strength.

**Desert Sand Fixation in Extreme Arid Environments.** Researchers in the Taklamakan Desert used *Sporosarcina pasteurii* to induce calcium carbonate crust formation, validating MICP's sand fixation effectiveness in extreme arid environments. Another study explored PCM (Phase Change Material)-assisted MICP technology for thermodynamic optimization and engineering application in desert soils under extreme conditions, providing innovative solutions for MICP

operation in high-temperature desert environments (daytime surface temperatures can reach 60–80°C).

**Effect of Different Calcium Sources on Solidification Performance.** A study on the effects of different calcium ion sources on MICP solidification in the Kashi Desert of Xinjiang showed significant differences among calcium sources (calcium chloride, calcium acetate, etc.). The unique advantage of the "Jianggang Plan" lies in the fact that f-CaO in steel slag, after organic acid activation and MICP catalysis, achieves in-situ sustained release of  $\text{Ca}^{2+}$  — this not only provides a continuous and stable calcium source for MICP solidification of desert sand but also solves the problems of high cost and uneven distribution of exogenous calcium salts.

#### **4.4 Module 4: Pilot Case — Successful Validation of the "Chinese Solution" in Middle Eastern Deserts**

In the Abu Dhabi desertification control demonstration project, a Chinese technical team used independently developed SBT (Special Bio-based Material) in a 1,750 m<sup>2</sup> desert experimental plot, successfully converting desert into fertile farmland. The water and nutrient retention rate of the improved soil jumped from below 5% to over 50%, pH approached neutral, and organic matter content was significantly improved. During the 120-day experimental period, 12 types of crops were successfully planted. Alfalfa achieved 7 harvests per year, radish was harvested twice, white radish yield per unit area reached 1.5 times that of local soil cultivation levels, corn and potato yields also exceeded local levels, and wheat yield reached 475 kg per mu ( $\approx 7,125$  kg/ha) in the first year.

Although this case did not directly use steel slag, its technical pathway is highly isomorphic with the "Jianggang Plan" — preparing soil improvement materials from biomass, constructing water-retaining structures in the desert, and activating soil microbial activity. Combining the successful experience of SBT with the mineral skeleton advantages of steel slag will provide solid technical accumulation for the implementation of the "Jianggang Plan" in the Middle East.

### **V. Five-Stage Ecological Succession Model (Desert Edition)**



Based on the "Five-Year, Three-Stage" core philosophy of the "Jianggang Plan," combined with the particularities of the desert environment and the feasibility verification of the technical modules described above, the following desert-edition five-stage ecological succession model has been designed:

### **5.1 Stage O: Site Selection and Substrate Preparation (Construction Period, 2–3 Months)**

**Core Tasks:** Select a desert plot near steel slag sources (e.g., JSIS Sohar Port facilities) and organic waste sources (e.g., Barka waste treatment plant). After clearing surface debris, alternately lay "steel slag layer → organic mixture layer → biochar layer → desert sand layer" to construct a 2–3 m deep multi-layer "Jianggang" reactor. Install an HDPE geomembrane and leachate collection system at the base. Surround the pile with date palm branches or straw as insulation layer. Install simple shade netting above the pile to reduce water evaporation during peak high-temperature periods.

### **5.2 Stage 1: Activation Period (Years 1–2)**

**Dominant Processes:** Rapid hydrolysis and acidification of organic waste, substantial accumulation of organic acids, local pH dropping from the original 12–13 of steel slag to a neutral-to-slightly-alkaline range of 6–8. Black soldier flies are inoculated and proliferate rapidly (initial stocking density: 5,000 individuals/m<sup>2</sup>, supplemented with 2,000–3,000 individuals/m<sup>2</sup> after 2 weeks), rapidly decomposing organic materials such as food waste. Organic acids carry out "in-situ etching" of the inert hydration layer on the surface of f-CaO in steel slag, releasing Ca<sup>2+</sup> into the solution phase in large quantities. Under high-temperature conditions (pile self-heating to 50–70°C), steel slag self-pulverization initiates. MICP microbial community (*Sporosarcina pasteurii*, CA-producing bacteria) is inoculated in batches, with initial inoculation accounting for 50% of the total. The first batch of fine CaCO<sub>3</sub> crystals (1–10 μm particle size) begins to precipitate on sand particle surfaces and steel slag particle surfaces.

**Desert-Specific Management:** Precision water supply using drip irrigation or subsurface irrigation systems, utilizing Treated Sewage Effluent (TSE) as the primary water source. Pile moisture is maintained through pulsed sprinkling (10–15 minutes every 6–12 hours), keeping the pile moisture content within the optimal range of 50–60%.

### **5.3 Stage 2: Stabilization Period (Years 3–4)**

**Dominant Processes:** The MICP microbial community becomes the dominant functional microorganism in the pile. CO<sub>2</sub> supply gradually transitions from organic matter degradation release to atmospheric CO<sub>2</sub> diffusion and, where available, industrial flue gas injection. CaCO<sub>3</sub> precipitation gradually grows from fine crystals into interconnected crystal clusters, and C-S-H gel begins filling pores between sand particles and steel slag particles. Pile temperature and pH gradually stabilize (pH 8–10), approaching natural desert conditions. Desert pioneer drought-tolerant plants (e.g., *Suaeda*, *Alhagi sparsifolia*, *Atriplex* spp.) begin natural colonization on the pile surface. Porosity decreases from 35–45% in Stage 1 to 25–35%, connected porosity declines, and water retention capacity continues to increase.

**Desert-Specific Management:** Supplementary MICP inoculant applied every six months (10% of total initial dose). 16S rRNA high-throughput sequencing conducted quarterly to monitor community structure. Salinization prevention and control: 10–15% of total leachate volume discharged monthly, supplemented by freshwater dilution from rainwater collection systems. Plant colonization employs "nurse plant" technology — preferentially planting nitrogen-fixing shrubs to create a micro-environment for subsequent crops.

### **5.4 Stage 3: Mineralization and Initial Ecological Park Formation (Year 5)**

**Dominant Processes:** The pile develops a stratified vertical structure. The surface layer (0–30 cm) has developed into a soil-like layer, with organic matter content reaching 2–5%. Plant and soil fauna communities are stabilizing, forming a preliminary desert oasis ecological landscape. The middle layer (30–100 cm) is a transition zone where carbonate cementation is evident; sand and steel slag particles

have been partially or completely cemented by the  $\text{CaCO}_3$  matrix. The deep layer (below 100 cm) is the mineralization zone, where f-CaO has been largely converted to  $\text{CaCO}_3$  and minor amounts of C-S-H gel. Porosity further decreases to 20–30%, forming a microstructure dominated by calcite pore-filling cementation supplemented by C-S-H gel matrix cementation.

**Desert-Specific Management:** Zonal rotational mining is conducted in the "artificial ore" extraction area, while the surface ecological zone is preserved as a permanent oasis. After homogenization, the extracted ore is prioritized for local construction materials (carbonated bricks, autoclaved bricks), followed by use as concrete aggregate.

#### **5.5 Stage 4: Ecological Park Maturity and Scale-up Replication (Year 6 and Beyond)**

**Core Tasks:** The first "Ecological Mine-Park" enters stable operation, with gradual reduction of manual intervention. Standard operating procedures for the "Desert Jianggang" are compiled and replicated across target regions (other provinces of Oman, UAE, Saudi Arabia, Mauritania, etc.). A "carbon sink monitoring–carbon trading" system is established, with applications submitted for VCS or GS international voluntary carbon reduction standard certification. The "Desert Jianggang" is promoted for inclusion in the host country's national ecological restoration plans and INDCs (Intended Nationally Determined Contributions).

## **VI. Overall Feasibility Assessment and Action Roadmap**

### **6.1 Comprehensive Assessment of Key Technology Module Applicability**

### **6.2 Key Scientific Questions (Pending Experimental Verification)**

Although the independent theoretical foundations of each technical module are already very solid, the following multi-factor coupling issues require dedicated experimental validation in desert environments:

**MICP Mineralization Efficiency Under High Temperature and Low Humidity:** Desert surface temperatures can reach 60–80°C with extremely high evaporation

rates. The synergistic regulation of temperature–humidity–microbial activity within the "Jianggang" pile needs validation. Countermeasures can be designed with reference to the latest research on PCM-assisted MICP technology.

**Impermeability Requirements Under Desert Groundwater Hydrological Conditions:** Desert groundwater tables are deep, but once contaminated, remediation is extremely difficult. The long-term reliability of the impermeable system requires specialized design.

**Long-term Management of Salinization and Salt Accumulation:** In desert environments where evaporation far exceeds precipitation, salt accumulation is a key challenge requiring focused attention in long-term operation. The long-term effectiveness of combined strategies — periodic discharge + rainwater dilution + halophyte absorption — needs validation.

**Multi-factor Coupled Dynamics Modeling for Desert Field Conditions:** Laboratory-scale (1–10 kg level) multi-factor coupled experiments on the desert sand–steel slag–organic waste mixed system, to establish a mathematical model of f-CaO transformation kinetics under desert conditions, represent the most critical scientific preparatory work before initiating a proof-of-concept engineering project.

### **6.3 Phased Action Roadmap**

## **VII. Conclusion**

After systematic validation, the overall conclusion of this report is as follows: The application of the "Jianggang Plan" for ecological restoration in desert regions of the Middle East (Oman, UAE) and Africa (Mauritania, etc.) possesses ample theoretical feasibility and pilot case support. It is recommended that the next phase of research be initiated as soon as possible.

Core justifications are as follows:

**Clear and Urgent Demand.** The MENA region suffers severe land degradation (44% of Oman's agricultural land affected by salinization). Desertification in Africa's

**Sahel region is a global focus, with FAO investing US\$222 million to support restoration. These regions are in urgent need of large-scale, low-cost, and sustainable ecological restoration solutions.**

**High Resource Matching.** Oman has abundant steel slag resources (existing JSIS facilities + large steel projects under construction) and organic waste (3,000 t/d municipal solid waste), extensive desert land, and a favorable policy environment (Vision 2040). African target regions have policy support frameworks (the "Great Green Wall," FAO SURAGGWA program) and a foundation of China-Africa cooperation (CAS Xinjiang Institute Mauritania project).

**Solid Theoretical Support for Each Technical Module.** Steel slag as a soil amendment has been field-validated in multiple countries (yield increases of 20%–200%, heavy metal risks controllable). Biochar can improve desert soil water use efficiency by 148%. The multi-scale mechanism of MICP desert sand fixation has been elucidated. The Abu Dhabi Chinese solution has achieved desert-to-farmland conversion (water retention rate jumping from below 5% to over 50%). The independent feasibility of each module has been fully validated.

**Key Risks Are Controllable.** Heavy metal risks can be controlled through a triple approach: self-stabilization mechanism + biochar immobilization + continuous monitoring. High-temperature low-humidity challenges can be addressed through PCM-assisted MICP technology and pile insulation design. Salinization accumulation can be managed through a combined strategy of periodic discharge + freshwater dilution + halophyte absorption.

**Significant Strategic Importance.** This plan systematically integrates China's steel slag resource utilization technology, MICP technology, and desert management experience. It represents an export of China's "national soft power" in the fields of ecological restoration and circular economy, perfectly aligning with the "Belt and Road" Green Silk Road Initiative and the framework of the China-Arab States Cooperation Forum.

**Recommended Actions to Be Initiated Immediately:**

**Within Two Weeks:** Send cooperation invitation letters to JSIS (Oman), Sultan Qaboos University, CAS Xinjiang Institute of Ecology and Geography, and other institutions to establish preliminary contact.

**Within One Month:** Complete the preparation of a laboratory-scale research plan for desert environment conditions (high-temperature low-humidity MICP mineralization efficiency, multi-factor coupled dynamics of steel slag–desert sand–organic waste).

**Within Two Months:** Organize an online/offline expert symposium, inviting experts from institutions described in this report — including KAUST, SQU, and the Xinjiang Institute of Ecology and Geography — to further refine and validate the technical approach.

**Within Six Months:** Complete funding applications to the FAO, Green Climate Fund (GCF), or China's "Belt and Road" Green Development Fund.

**Expert Team Support Acknowledgment:** This report was compiled with full integration of the academic expertise of the original eight-member "Jianggang Plan" expert panel, and with reference to the pioneering research findings of Professor Himanshu Mishra's team at KAUST (organic waste biochar), Dr. Ahmed Rashid Al-Rubaiai's team at SQU (sulfur-treated biochar for saline-alkali soil improvement), the Xinjiang Institute of Ecology and Geography, CAS (Mauritania desertification control project), and GreenTech (Shanghai) Agricultural Technology Co., Ltd. (Abu Dhabi SBT desert improvement project). Deep respect and gratitude are hereby extended.

| Resource Element  | Oman                                  | UAE            | Saudi Arabia          | Mauritania                        |
|-------------------|---------------------------------------|----------------|-----------------------|-----------------------------------|
| Steel slag source | ★★★★★<br>JSIS existing + new projects | ★★★<br>Limited | ★★★★★<br>Hadeed, etc. | ★★<br>Needs import or alternative |
| Organic waste     | ★★★★                                  | ★★★★           | ★★★★                  | ★★★                               |

|                              |                              |                               |                                |                                    |
|------------------------------|------------------------------|-------------------------------|--------------------------------|------------------------------------|
|                              | 3,000 t/d<br>municipal waste | Municipal + agri-<br>residues | Date palm +<br>municipal waste | Agri-residues +<br>municipal waste |
| Desert land                  | ★★★★★<br>Vast areas          | ★★★★★<br>Vast areas           | ★★★★★<br>Vast areas            | ★★★★★<br>Vast areas                |
| Policy support               | ★★★★★<br>Vision 2040         | ★★★★★<br>Green Agenda         | ★★★★★<br>Green Agenda          | ★★★★★<br>Great Green Wall          |
| Water resources              | ★★<br>Limited                | ★★<br>Limited                 | ★★<br>Limited                  | ★★<br>Limited                      |
| Comprehensive<br>feasibility | Very High                    | High                          | High                           | Medium-High                        |

| <b>Technology<br/>Module</b>   | <b>Desert<br/>Applicability</b> | <b>Theoretical<br/>Support</b>                               | <b>Pilot Cases</b>                                                  | <b>Key Risks &amp;<br/>Mitigation</b>                                                   |
|--------------------------------|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Steel slag soil<br>improvement | ★★★★★                           | Very strong<br>(multi-country 20-<br>200% yield<br>increase) | Korea,<br>Bangladesh,<br>Japan;<br>Japan/Europe slag<br>fertilizers | Cr risk → self-<br>stabilization +<br>biochar<br>immobilization +<br>monthly monitoring |
| Biochar water<br>retention     | ★★★★★                           | Very strong<br>(WUE +148%,<br>yield +146%)                   | NE Brazil semi-<br>arid;<br>Oman native<br>saline-alkali            | Feedstock → local<br>date palm<br>residues, agri-<br>residues                           |
| MICP sand<br>fixation          | ★★★★★                           | Very strong<br>(macro-to-micro<br>mechanism)                 | Taklamakan<br>Desert;<br>lab optimization<br>validated              | High temp → PCM-<br>assisted;<br>Ca source → steel<br>slag slow-release                 |
| Black soldier fly              | ★★★★                            | Strong<br>(desert project<br>attempts)                       | Edama Saudi<br>Arabia<br>100% organic<br>recovery                   | Temp limits →<br>insulation +<br>seasonal rotation                                      |
| Multi-factor<br>synergy        | ★★★★★                           | Strong<br>(biochemical<br>coupling<br>elucidated)            | Abu Dhabi SBT<br>closest<br>precedent case                          | Holistic validation<br>→<br>bench→pilot→demo                                            |

| Stage                              | Timeframe    | Objective                       | Core Tasks                                                                                    | Est. Investment    |
|------------------------------------|--------------|---------------------------------|-----------------------------------------------------------------------------------------------|--------------------|
| Stage O:<br>Preliminary<br>Liaison | Months 1–3   | Establish cooperation network   | Contact JSIS, SQU, UAE GreenTech, Mauritania Green Wall, CAS Xinjiang; apply to FAO SURAGGWA  | Travel & personnel |
| Stage 1:<br>Lab-scale              | Months 4–15  | Validate desert parameters      | Simulate high-temp low-humidity; multi-factor coupled experiments; desert f-CaO kinetic model | RMB 0.5–1M         |
| Stage 2:<br>Field pilot            | Months 16–30 | Desert field validation         | 100–500 t pilot piles; mass-heat-reaction field coupling; test local materials                | RMB 2–5M           |
| Stage 3:<br>Proof-of-concept       | Months 31–54 | 10,000 t engineering validation | 10,000 t Desert Jianggang PoC; artificial ore brick pilot; life-cycle carbon footprint        | RMB 20–50M         |
| Stage 4:<br>Scale-up               | Month 55+    | Industrialization               | Technical standards; regional replication; carbon credit trading                              | Project financing  |